Business Calculus (MATH 2350)
Summer 2018
Friday, 06/22/2018
Quiz 7: Section 12.5, 12.6

Name (Print): ______________________

Score: / 100

Section 12.5

1. $\boxed{\text{15 points}}$ Let $f(x) = x^2 - 2$. Find the absolute minimum and the absolute maximum of $f(x)$ over the interval $[-5, 5]$, if such an absolute maximum or minimum exists. For every absolute extremum, give the point $(x, f(x))$ and indicate whether it is a minimum or maximum.

2. $\boxed{\text{15 points}}$ Let $f(x) = x^2 - 2$. Find the absolute minimum and the absolute maximum of $f(x)$ over the interval $[2, 3]$, if such an absolute maximum or minimum exists. For every absolute extremum, give the point $(x, f(x))$ and indicate whether it is a minimum or maximum.

3. $\boxed{\text{15 points}}$ Let $f(x) = ln(x) - \frac{1}{2}x$. Find the absolute minimum and the absolute maximum of $f(x)$ over the interval $(1, 20)$, if such an absolute maximum or minimum exists. For every absolute extremum, give the point $(x, f(x))$ and indicate whether it is a minimum or maximum.

4. $\boxed{\text{15 points}}$ Let $f(x) = x^3 - 3x^2 + 1$. Find the absolute minimum and the absolute maximum of $f(x)$ over the interval $[1, 15]$, if such an absolute maximum or minimum exists. For every absolute extremum, give the point $(x, f(x))$ and indicate whether it is a minimum or maximum.

5. $\boxed{\text{15 points (bonus)}}$ Give an example of a function $f(x)$ for which $(x, f(x))$ is both an absolute maximum and an absolute minimum for every x in the interval $(-\infty, \infty)$.

Section 12.6

1. $\boxed{\text{20 points}}$ A coffee shop sells 800 cups of coffee per week at a price of \$4 each. A market survey shows that for every \$0.01 reduction in price, 3 more cups will be sold. How much should the coffee shop charge for coffee in order to maximize revenue?

2. $\boxed{\text{20 points}}$ A box is to be made at a manufacturing company out of a square piece of cardboard 10 inches wide. Squares of equal size will be cut out of each corner, and then the ends and sides will be folded up to form a rectangular box with an open top. What size square should be cut from each corner to maximize volume?